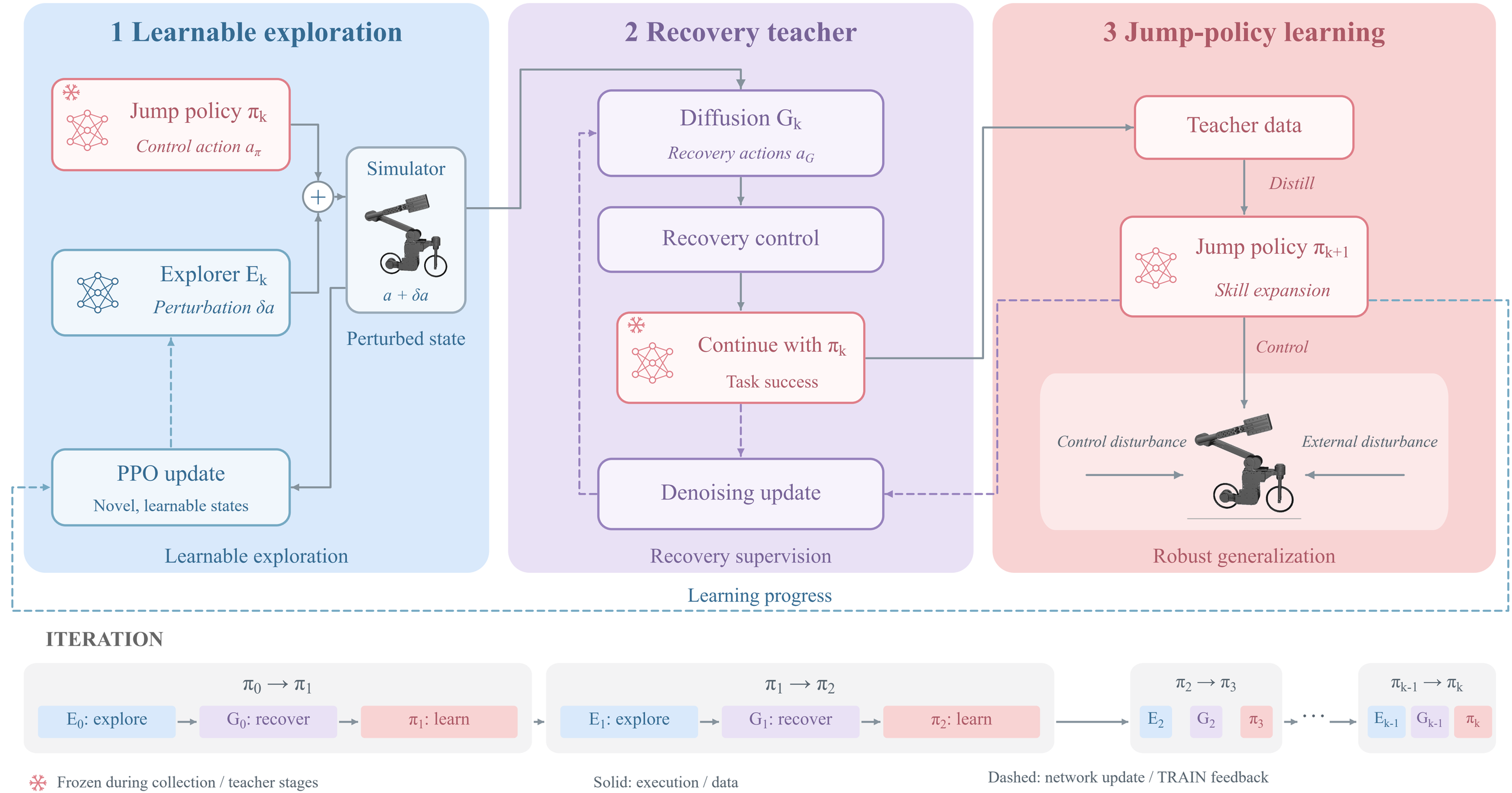

1 Learnable exploration
Jump policy π_k
Control action a_π
Explorer E_k
Perturbation δa
Simulator
$a + \delta a$
Perturbed state
PPO update
Novel, learnable states
Learnable exploration
2 Recovery teacher
Diffusion G_k
Recovery actions a_G
Recovery control
Continue with π_k
Task success
Denoising update
Recovery supervision
3 Jump-policy learning
Teacher data
Distill
Jump policy π_{k+1}
Skill expansion
Control
Control disturbance
External disturbance
Robust generalization
Learning progress
ITERATION
$\pi_0 \rightarrow \pi_1$
E_0: explore
G_0: recover
π_1: learn
$\pi_1 \rightarrow \pi_2$
E_1: explore
G_1: recover
π_2: learn
$\pi_2 \rightarrow \pi_3$
E_2
G_2
π_3
$\pi_{k-1} \rightarrow \pi_k$
E_{k-1}
G_{k-1}
π_k
Frozen during collection / teacher stages
Solid: execution / data
Dashed: network update / TRAIN feedback